

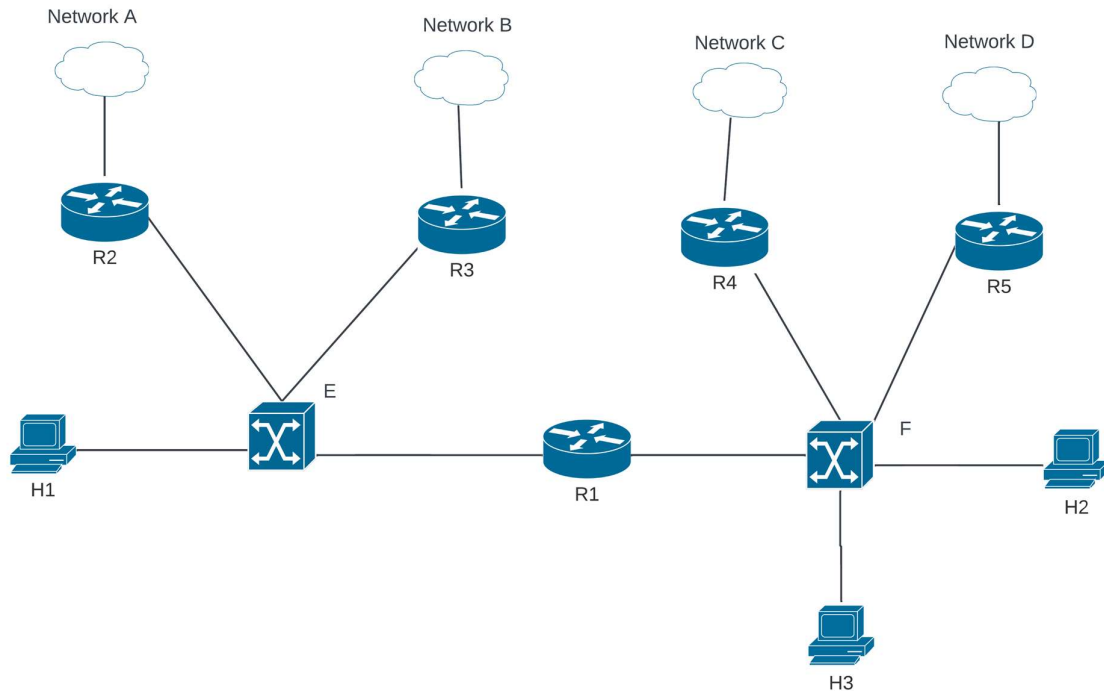
EP2120 Internetworking IK2218 Protocols and Principles of the Internet

Homework Assignment 1 **(Solutions due 19:00, Mon., 7 Sept. 2026)** **(Review due 19:00, Wed., 9 Sept. 2026)**

1. IPv4 Addressing (30/100)

- What is the best-fit netmask (i.e., resulting in as few unused host addresses as possible) for a network with 25 hosts and 2 routers connected to it? If the subnet provides more IP addresses than required, calculate the number of unused IP addresses. (5p)
- b) Assume that the network 10.20.0.0/22 must be divided into equal-sized subnets, each supporting at least 50 usable IP addresses for hosts and routers. What is the maximum number of equal-sized subnets that can be created? What is the total number of IP addresses available for hosts and routers across all resulting subnets? (5p)
 - c) The network 172.16.40.0/24 is divided into four subnets that need to support 96, 60, 28, and 20 usable IP addresses. Assign IP address blocks to the four subnets in ascending order of size. For each subnet, provide: the network address in binary notation, the corresponding netmask in binary form, and the CIDR notation. (5p)
 - d) What is the directed broadcast address of the network 172.20.4.160/27 (5p)
 - e) Consider a host with IP address 172.16.75.142. The host is in a subnet with directed broadcast address 172.16.75.255. What is the network address of the largest and smallest possible subnet that the host may belong to? Provide the results in dotted decimal notation, including the prefix length. (5p)
 - f) Use the services of IANA to identify the appropriate Regional Internet Registry (RIR), and use the appropriate RIR to figure out to whom the IP address block 194.71.64.0/22 belongs. Provide the name of the organization and the AS number. (5p)

2. Address allocation (30/100)



Consider the network above, a routed network in an organization's enterprise network. Network E is a switched Ethernet network containing R1, R2, R3, and monitoring host H1. Network F is a separate switched Ethernet network containing R1, R4, R5, and monitoring hosts H2 and H3. Routers R2, R3, R4, and R5 connect to local office networks A, B, C, and D, respectively. R1 provides connectivity to the rest of the Internet. All networks use Ethernet.

The organization wants to make an *address allocation* by assigning an address block to networks A to F in the following way:

- Network A contains 300 hosts, network B contains 126 hosts, and networks C contains 500 hosts, and network D contains 510 hosts (excluding the router interfaces). The lowest address should be assigned to network D.
- There are no unnumbered point-to-point links: all Ethernet networks are IP subnets and all nodes (routers and hosts) have an IP address on all their network interfaces. All nodes need to be reachable from any other node.
- The address allocation should be such that the subnets can be *aggregated* as much as possible.
- Each subnet should not be larger than necessary to accommodate all nodes (host or router interfaces) in the subnet. The number of unassigned IP addresses should be minimal.

a) Assume that the address block that is available for address allocation is 10.88.0.0/x. Find all possible values of the prefix length x for which an address allocation that satisfies the above requirements is possible. Provide a justification for the solution. (8p)

b) Consider now the maximum value of x you found in (a). Make an address allocation as described in (i-iv). Based on your address allocation, provide the address blocks assigned to networks A to F in CIDR notation. Give a sketch of your reasoning to support your solution. (15p)

c) Provide the required entries of the forwarding table of router R1 for networks A-F (ignore the default route toward the rest of the network). Choose appropriate IP addresses for the router interfaces that can appear as next hop in the forwarding table. (7p)

3. IPv4 forwarding (20/100)

A network administrator is investigating a potential problem with a router. The network administrator runs a series of tests to identify the problem and observes that datagrams with the below destination addresses are successfully delivered to their destinations:

- a) 193.147.1.31
- b) 10.64.2.1
- c) 190.1.45.129
- d) 144.127.32.211

Later, the network administrator realizes that interface Eth2 of the router was down. The forwarding table of the router is given below. The destination column contains the network address and the prefix length, as usual. Missing entries are represented using X, Y, Z, and T.

Destination	Next hop	Flags	Interface
10.X.0.0/16	-	U	Eth0
144.Y.0.0/5	10.64.2.3	UG	Eth0
190.1.45.Z/29	10.64.1.6	UG	Eth0
150.2.0.0/16	-	U	Eth1
193.147.1.0/T	150.2.3.1	UG	Eth1
0.0.0.0/0	10.34.2.129	G	Eth2

Given the information available to the network administrator what could be the possible values for X, Y, Z, and T? Motivate your answer.

4. IPv4 and IPv6 datagram formats (20/100)

An IP datagram has arrived to a network interface with the following data in the IP header (in hexadecimal format):

45 00 00 3C 1C 46 20 B9 40 06 51 86 C0 00 02 01 C6 33 64 02

- a. What is the protocol version? What is the type of service (also called the DS field)?
- b. Is this IP datagram a fragment? Why?
- c. What is the size of the IP payload?
- d. Do we know if the IP header was corrupted during transmission? What would happen to the datagram if the checksum does not match?
- e. How many routers has the packet traversed before we observed it, assuming the sending host is running Ubuntu (Hint: check the default TTL for Ubuntu Linux)?
- f. What is the identification number of the datagram? What is it used for?

- g. What is the source address in dotted decimal format? Is it routable?
- h. What is the destination address in dotted decimal format? Is it routable?
- i. Which transport layer protocol is used, and how can you tell? Is the transport layer header included in the payload of this datagram?
- j. Are there any IPv4 options used?

(Each question is worth 2 points.)